Operating Systems: Concurrency, Synchronization & Memory

Chapter 1: Process Management & CPU Scheduling Algorithms

A process is an instance of a computer program in execution, consisting of program code, current activity represented by the program counter, stack, data section, and heap memory. In contrast, a thread is a lightweight unit of CPU utilization that shares address space with peer threads.

- **Process Control Block (PCB):** Stores PID, Process State (New, Ready, Running, Waiting, Terminated), CPU registers, and I/O status.
- **Context Switch:** Saving the state of the active process and restoring the state of the scheduled process; pure operational overhead.
- **First-Come, First-Served (FCFS):** Non-preemptive scheduling prone to the convoy effect where short processes wait behind long CPU bursts.
- **Shortest Job First (SJF):** Mathematically optimal for minimum average waiting time; requires burst time prediction.
- **Round Robin (RR):** Preemptive time-sliced scheduling with quantum q; optimal for interactive time-sharing systems.

Chapter 2: Concurrency, Critical Sections & Deadlock Avoidance

When concurrent processes share resources (such as memory or files), race conditions occur if the outcome depends on the order of execution. The Critical Section Problem requires Mutual Exclusion, Progress, and Bounded Waiting.

- **Mutex Locks:** Binary synchronization primitives utilizing acquire() and release() operations with atomic test-and-set instructions.
- **Semaphores:** Integer variables accessed through atomic wait(S) [P] and signal(S) [V] functions; counting semaphores manage finite resource pools.
- **Deadlock Coffman Conditions:** Mutual Exclusion, Hold and Wait, No Preemption, and Circular Wait.
- **Banker's Algorithm:** Deadlock avoidance strategy. State is safe if Need matrix <= Available vector: Need[i][j] = Max[i][j] - Allocation[i][j].
- **Indian Industry Application (Zerodha Kite):** Low-latency lock-free ring buffers in memory to broadcast real-time stock quotes to millions of traders without deadlocks.

Chapter 3: Virtual Memory Management & Page Replacement

Virtual memory decouples the user logical memory from physical memory, permitting the execution of processes that require larger memory spaces than physically installed RAM.

- **Paging Mechanism:** Logical address space is divided into fixed-size Pages; physical memory into Frames. Handled by Memory Management Unit (MMU) with Translation Lookaside Buffer (TLB).
- **Page Fault Rate Formula:** Effective Access Time (EAT) = (1 - p) * ma + p * (Page Fault Overhead), where p is probability of page fault and ma is memory access time.

• **Least Recently Used (LRU):** Replaces the frame that has not been accessed for the longest period of time; obeys Belady's anomaly protection.

• **Thrashing:** High paging activity where CPU spends more time swapping pages in/out of disk than executing user instructions.